Capstone Project

End-to-end AI on an unseen dataset, written up as a detailed report · 40% of the module · LO3, LO4, LO5

1. What you will do

This is the module's one capstone project. You will take a dataset you have never seen, and that general-purpose AI assistants have never seen, through a complete project: understand the data, build and train a model, write the core of the learning machinery yourself, and audit what you built. You then write it all up in a **detailed report** that contains every result, the links to every other deliverable, and the details of how you produced them. The report is submitted by 6 January 2027 and carries 90% of the marks; a 10-minute in-person interview in the week of 11 January 2027 carries the other 10%. You work in one of six electives; each is a different route through the same brief, with the same deliverables and the same marking.

2. Your elective

	Elective	Your data
E1	Supervised Learning with Tabular Health Data	De-identified patient records; predict who returns to hospital within 30 days
E2	Unsupervised Learning with Agricultural Data	Unlabelled field and sensor records; find management zones and show they are real
E3	Reinforcement Learning or Transformers for Textual Data	Recent short messages with routing labels; a simulated reward environment for the reinforcement-learning route
E4	Transfer Learning for Buying a House	Recent property listings with details, photographs and descriptions; predict a fair price
E5	Physics-Informed Machine Learning, with No Data	None given: you generate data from the equations of your own variant of a cooling system
E6	Physics-Informed Neural Networks, with No Data	None at all: your own variant of a partial differential equation is the only teacher

The data for E1 to E4 is released by the teaching team in Week 9 and is not public. For E5 and E6 your personal variant, generated from your student number, is released at the same time, so no two projects share an answer. In Week 8 you submit a short proposal through the **Capstone Proposal Form** linked on Tutors (Capstone Project · Your Proposal); your answers come back in Week 9 as your written proposal and your week-by-week implementation plan, reviewed by the lecturers.

3. Deliverables

	Deliverable	What it must contain
D1	Report	A detailed report written in LaTeX or Word from the course template and submitted as a PDF: all your results, the links to every other deliverable, and the details of how you produced them (section 4).
D2	Repository	Your code, with a documented commit history from early in the project and a README that reproduces your results from a clean clone with one command.
D3	Logbook	A short record of decisions, experiments, dead ends and fixes, written as you go, including how you used AI tools.
D4	Trained model	A model trained on your assigned data, with real loss and metric curves.
D5	From-scratch components	Backpropagation, an optimiser step and PCA via SVD written by you, each with tests against a reference implementation.
D6	Ethics and fairness audit	Measured group metrics on your own model, one mitigation, and its measured cost. In E5 and E6, reliability across operating regimes or across the domain instead of group metrics.
D7	Explainability summary	SHAP or LIME (or, in E5 and E6, learned physical parameters or residual maps) showing what drives your model's decisions.
D8	Model card and governance note	What the model is for, how it performs for whom, its limits, and the rules that apply to it.
D9	In-person interview (week of 11 January 2027)	Ten minutes with both lecturers: you show one key result running from your repository, then answer questions on your report, code, audit and logbook.

4. Your report

Write the report in LaTeX or Word from the course template (on Tutors, and in `report/` in the project starter); both have the same outline, tables and appendices. The main body is about 5,000 words (between 4,000 and 6,000), not counting the title page, abstract, references and appendices.

	Section	Length	Marked under
	Title page and abstract	150 to 200 words	M6
1	Introduction: the question, who it is for, what you built	400 to 600	M1
2	Data: source, profile, preparation, split and leakage	600 to 900	M1, M2
3	Methods: baseline, main model, training set-up, from-scratch components and their tests	800 to 1,100	M1, M3
4	Results: headline table with spread, curves, error analysis, experiments	900 to 1,300	M2
5	Responsibility: audit, mitigation and its cost, explanations, model card and governance	700 to 1,000	M4
6	Discussion: meaning, limitations, dead ends, next steps	400 to 700	M1, M2, M4
7	Conclusion	150 to 250	M6
	References; Appendices A (deliverables and links), B (reproducibility), C (test results), D (use of AI tools), E (changes from the proposal)		M2, M3, M5, M6

Submit by 23:59 on Wednesday 6 January 2027: the report PDF on Moodle, and the commit tagged `final` (with the same PDF and its source in `report/`) pushed to your repository.

5. Milestones

When	What	How it helps
Week 8	Capstone project introduced; proposal form submitted	Fixes your elective, your question and your plan.
Week 9	Proposal returned; data (or personal variant) released; data understood; baseline trained	Follows your implementation plan.
Week 10	Main model trained; from-scratch components passing their tests	Evidence for M2 and M3.
Week 11	Audit, explainability, model card and governance note	Evidence for M4.
Week 12 to 6 January 2027	Report written from the template; repository checked from a clean clone; report PDF on Moodle and commit tagged `final` by 23:59 on 6 January 2027	Evidence for M1 to M6.
Week of 11 January 2027	10-minute in-person interview (D9)	M7.

6. How it is marked

Element	Weight within the project	Learning outcomes
M1 · Report: question, data and approach	15%	LO3 (and LO2)
M2 · Report: results, reproducibility and correctness	25%	LO3
M3 · Report: from-scratch components and their tests	15%	LO3 (and LO1)
M4 · Report: ethics, fairness and explainability audit	15%	LO5
M5 · Report: process evidence (repository history, log-book, AI use)	10%	LO3
M6 · Report: clarity, structure and completeness	10%	LO3, LO5
M7 · In-person interview (D9)	10%	LO3, LO4, LO5

Each element is marked out of 100 against the rubric in document 05. The report carries 90% of your project mark and the interview 10%; your project mark counts 40% towards the module, and it is the only project in the module.

7. Using AI tools

- You may use AI tools to help you, as a helper and not a replacement.
- Record every use in your logbook (what you asked, what you used, what you changed), and summarise it in Appendix D of your report.
- You must be able to explain every line you submit; the interview will ask you to, including your from-scratch code.